

Empirical Proof Record: VALIDATED

2026-05-24T03:29:09.049571+00:00

Empirical Proof Record — VALIDATED

Proof ID: 4cefa5ac4f69ccd8f7660679ec2bde55...

Created: 2026-05-24T03:29:09.049571+00:00

1. Claim

The posterior 94% HDI width on μ_Φ contracts at the theoretical $\sqrt{N}$ rate as the number of observations grows. Specifically, $\text{HDI_width}(N=200) / \text{HDI_width}(N=20) \leq 0.40$ (theoretical contraction factor $\sqrt{(20/200)} = 0.316$; the 0.40 ceiling allows finite-sample slack).

- **Operationalisation:** Fit a PyMC NUTS posterior to subsamples of the Φ trajectory at sizes 20, 50, 100, 200. Posterior model: $\mu \sim \text{Normal}(\text{prior})$, $\sigma \sim \text{HalfNormal}$, $y_i \sim \text{Normal}(\mu, \sigma)$. Compute the 94% HDI width on μ at each size; the contraction metric is $\text{HDI_width}(N=200) / \text{HDI_width}(N=20)$.
- **Threshold:** `hdi_contraction_ratio <= 0.4 ratio`
- **H0:** `hdi_contraction_ratio > 0.40` (posterior fails to tighten at the predicted rate — data may be non-Gaussian, non-stationary, or sampler miscalibrated)
- **H1:** `hdi_contraction_ratio <= 0.40` (posterior tightens approximately at the theoretical $\sqrt{N}$ rate, confirming Bayesian inference behaves correctly on the Φ trajectory)

2. Verdict

VALIDATED — observed 0.27735050659795846

HDI width $N=20 \rightarrow N=200$: 0.0654 $\rightarrow$ 0.0181; contraction ratio 0.2774 (theoretical 0.3162; ceiling 0.4000). Empirical Φ mean=0.6140 sd=0.0680 (n=200).

3. Evidence

Pillar	Statistic	Value	95% CI	Library
pymc_nuts_posterior	hdi_contraction_ratio	0.27735050659795846	—	pymc 6.0.0

4. Pre-registration

- Registered at: 2026-05-24T03:29:08.777203+00:00
- Config hash: a47e4fd070e7935308f57886af093806...
- Data hash: 9ef317fe291445319fcba522d29836dc...

5. Signature

2d2cd20e6cac4f3e55cd68d961f969d50731ec191d4815f948c13b986ce9bed3